# SARVARBEK ERNIYAZOV

Ph.D. Candidate, Computer Engineering | Reliable Learning and Decision-Making for Energy and Cyber-Physical Systems

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Google Scholar | GitHub | Research Portfolio | Hugging Face | LinkedIn

## Academic Profile

I develop and evaluate learning methods for heterogeneous energy and cyber-physical systems, with an emerging focus on decision reliability under distribution shift. My doctoral research advances spatio-temporal graph learning, multimodal mixture-of-experts, frequency-aware modeling, and federated learning for solar, building-energy, and industrial-control systems; my earlier research in low-power quantum-dot cellular automata established a hardware-aware perspective on energy, latency, and computational constraints. My independent direction asks when a distributed model should use a prediction, defer the decision, or request additional information as sensor and site conditions change.

## Research Focus

Spatio-temporal and multimodal learning for energy and cyber-physical systems • Federated and personalized learning under client heterogeneity • Uncertainty quantification, conformal risk control, and selective prediction • Distribution shift, anomaly detection, and failure-centered evaluation • Resource-aware AI for edge–cloud physical infrastructure.

## Education and Availability

**Ph.D. in Computer Engineering**, Chonnam National University, Yeosu, South Korea — 2020 – Feb. 2027 (exp.)

Advisor: Prof. Chang Gyoon Lim • Dissertation defense scheduled: **December 2026** • Official degree conferral: **February 2027**

**M.S. in Computer Engineering**, Kumoh National Institute of Technology, Gumi, South Korea — 2016 – 2018

Information Security Laboratory (QCA Research Group)

**B.S. in Computer Engineering (Information Security)**, Tashkent University of Information Technologies, Uzbekistan — 2011 – 2015

My background combines undergraduate information-security training, master's research on low-power QCA circuits within an information-security laboratory, and doctoral work on industrial-control anomaly detection.

**Postdoctoral availability:** from March 2027, and compatible with later start dates. My university affiliation and grant responsibilities conclude upon degree conferral in February 2027; the funded project continues to 2028, but my personal participation ends at conferral.

## Selected Research Impact

- **3 peer-reviewed AI journal articles** (2024–2026), all first-author; **9 peer-reviewed publications** in total with verified records (7 journal, 2 proceedings); one manuscript under review, listed separately.
- **124 citations** across the full research record, including **93** to the lead-author 2019 *Microelectronic Engineering* article (Google Scholar, 11 September 2026).
- **10 public research repositories** and **7 hosted research interfaces**; technical contributions to funded Korean national R&D in virtual power plants, edge–cloud energy management, and industrial monitoring.

## Representative Publications

1. **S. Erniyazov** and C. G. Lim. "GNN-enhanced temporal patch segmentation and frequency fusion model for robust solar energy production forecasting." *Energy Reports*, 13, 4962–4984, 2025. doi:10.1016/j.egyr.2025.03.063
   First-author; up to 46% lower reported MSE at 12 h versus PatchTST, Crossformer, PatchMixer, and Fredformer under the study protocol.
2. **S. Erniyazov**, M. A. Jaleel, C. G. Lim, and S. B. Ha. "A multimodal fusion framework for solar forecasting using dynamic GNNs and mixture of experts." *KSII Transactions on Internet and Information Systems*, 20(3), 1337–1360, 2026. doi:10.3837/tiis.2026.03.012
   First-author; dynamic graph and horizon-aware expert routing; 49% reported forecast skill relative to the NAM physical model.
3. **S. Erniyazov**, Y.-M. Kim, M. A. Jaleel, and C. G. Lim. "Comprehensive analysis and improved techniques for anomaly detection in time series data with autoencoder models." *International Journal on Advanced Science, Engineering and Information Technology*, 14(6), 2024. doi:10.18517/ijaseit.14.6.20716
   First-author; HAI industrial-control-system benchmark; 99% accuracy and F1 = 0.987 under the study protocol.
4. **S. Erniyazov** and J.-C. Jeon. "Carry save adder and carry look ahead adder using inverter chain based coplanar QCA full adder for low energy dissipation." *Microelectronic Engineering*, 211, 37–43, 2019. doi:10.1016/j.mee.2019.03.015

## Research Experience

**Ph.D. Researcher — AI Laboratory, Chonnam National University** 2020 – present
Department of Computer Engineering, Yeosu, South Korea • Advisor: Prof. Chang Gyoon Lim
**Research program:** spatio-temporal, multimodal, and distributed learning for energy forecasting and cyber-physical anomaly detection.

- Led first-author research on graph-enhanced temporal and frequency modeling for multi-station solar forecasting, including architecture design, implementation, baseline comparison, ablation analysis, and manuscript preparation.
- Developed dynamic graph and mixture-of-experts methods for integrating heterogeneous solar and meteorological modalities across forecast horizons.
- Investigated federated spatio-temporal mixture-of-experts for heterogeneous building-energy clients; the manuscript is under review and is excluded from the peer-reviewed publication count.
- Developed and evaluated autoencoder-based time-series anomaly detection on the HAI industrial-control-system benchmark.
- Built reproducible research pipelines: data validation, leakage-aware splitting, training, statistical evaluation, ablations, and research-prototype integration.
- Contributed AI methodology and technical deliverables to Korean national R&D programs involving virtual power plants, edge–cloud energy management, satellite–ground data integration, and power-infrastructure monitoring.

**Individual contribution statement:** For first-author doctoral studies, I led method development, software implementation, experimental design, baseline and ablation evaluation, result visualization, and manuscript preparation. Formal grant governance and institutional PI responsibilities remained with the appointed faculty principal investigator.

**Graduate Researcher — Information Security Laboratory / QCA Research Group** 2016 – 2018
Kumoh National Institute of Technology, Gumi, South Korea

- Designed and simulated low-power quantum-dot cellular automata circuits, including full adders, carry-save and carry-look-ahead adders, magnitude comparators, subtractors, and reversible logic.
- Reported more than 50% area reduction for the proposed carry-look-ahead architecture relative to the specified comparison design.
- Contributed to the laboratory's peer-reviewed QCA publication record, including the 2019 *Microelectronic Engineering* article.
- Conducted this work within an information-security laboratory, following undergraduate information-security training. The resulting perspective on energy, latency, sensing, and resource constraints now informs my cyber-physical AI research.

## National R&D Technical Leadership

**Role formulation used throughout:** Lead AI Architect / Lead Researcher for the assigned technical work package; not the formal grant Principal Investigator. I led the assigned AI work package from problem formulation and data-interface design through model architecture, leakage-aware validation, and research-prototype integration, coordinating technical interfaces with the faculty PI, domain partners, and software contributors while grant governance remained with the appointed PI.

### TIPA RS-2026-25529976 — sLM–LMM Cooperative Virtual Power Plant Platform

[Funded and active | 2026–2028] Industry–academia commercialization R&D for on-device AI and cloud-integrated energy management. **Technical role:** Lead AI Architect / Lead Researcher for the assigned university AI work package; not the formal grant PI.

- Define model and evaluation architecture for cooperation between on-device small language models and cloud-based multimodal models in VPP workflows.
- Lead AI-side interface design connecting telemetry, forecasts, operator tools, and edge–cloud inference components.
- Direct technical validation for assigned machine-learning deliverables and coordinate research-prototype integration with faculty and industry partners.

**Appointment clarification:** my university affiliation and associated project responsibilities conclude upon Ph.D. degree conferral in February 2027; no project, funding, or time commitment will continue into a postdoctoral appointment.

### TIPA RS-2025-02312851 — Preliminary Study for an On-Device AI and Cloud-Integrated VPP Platform

[Funded and completed | 2025] **Technical role:** Lead AI Researcher for the machine-learning feasibility work: led AI feasibility analysis, model architecture planning, and prototype evaluation supporting the subsequent commercialization-stage project, and translated research requirements into a technically scoped edge–cloud VPP development plan.

### Selected Proposal Development — Not Funded Awards

- **NRF RS-2025-00564485** [submitted / under review] — satellite–ground data integration for solar variability management and VPP operations; contributed multimodal mixture-of-experts and on-device-AI methodology.
- **INNOPOLIS RS-2026-25550513** [submitted / under review] — global validation of AI-based power-equipment anomaly detection and preventive control; contributed AI anomaly-detection and validation methodology.

Submitted proposals are listed separately and are not counted as funded grants.

## Publications

Verified publication records are listed below, with manuscripts under review reported separately and excluded from peer-reviewed publication counts. Citation counts were confirmed on Google Scholar on 11 September 2026. My name is shown in bold. Full record: Google Scholar and ORCID.

**Record summary** 7 peer-reviewed journal articles • 2 peer-reviewed proceedings papers • 1 manuscript under review • first author on every entry

### Peer-Reviewed Journal Articles — Published or Accepted

[J1] **S. Erniyazov**, M. A. Jaleel, C. G. Lim, and S. B. Ha. "A multimodal fusion framework for solar forecasting using dynamic GNNs and mixture of experts." *KSII Transactions on Internet and Information Systems*, 20(3), 1337–1360, 2026. doi:10.3837/tiis.2026.03.012
Contribution: first-author multimodal architecture, dynamic graph and expert-routing methodology, evaluation, analysis, and manuscript preparation.

[J2] **S. Erniyazov** and C. G. Lim. "GNN-enhanced temporal patch segmentation and frequency fusion model for robust solar energy production forecasting." *Energy Reports*, 13, 4962–4984, 2025. doi:10.1016/j.egyr.2025.03.063
Contribution: first-author method development, implementation, experimental design, baseline and ablation evaluation, and manuscript preparation.

[J3] **S. Erniyazov**, Y.-M. Kim, M. A. Jaleel, and C. G. Lim. "Comprehensive analysis and improved techniques for anomaly detection in time series data with autoencoder models." *International Journal on Advanced Science, Engineering and Information Technology*, 14(6), 2024. doi:10.18517/ijaseit.14.6.20716
Contribution: first-author model design, HAI benchmark evaluation, analysis, and manuscript preparation.

[J4] **S. Erniyazov** and J.-C. Jeon. "Carry save adder and carry look ahead adder using inverter chain based coplanar QCA full adder for low energy dissipation." *Microelectronic Engineering*, 211, 37–43, 2019. doi:10.1016/j.mee.2019.03.015
93 citations (Google Scholar, 11 September 2026). Contribution: circuit design, simulation, comparative evaluation, and manuscript preparation.

[J5] **S. Erniyazov** and J. C. Jeon. "Coplanar subtractor design in QCA for arithmetic circuit design." *Advanced Science Letters*, 23(10), 10112–10117, 2017. doi:10.1166/asl.2017.10399

[J6] **S. Erniyazov** and J. C. Jeon. "Implementation of area efficient Fredkin gate for compact reversible QCA circuit." *Advanced Science Letters*, 23(10), 10092–10096, 2017. doi:10.1166/asl.2017.10395

[J7] **S. Erniyazov** and J. C. Jeon. "Reversible circuit design in QCA based on double Feynman gate." *Advanced Science Letters*, 23(10), 9852–9856, 2017. doi:10.1166/asl.2017.9810

### Peer-Reviewed Conference or Proceedings Papers

[C1] **S. Erniyazov** and J.-C. Jeon. "Area efficient magnitude comparator based on QCA." *Advanced Science and Technology Letters*, 150, 75–79, 2018. doi:10.14257/astl.2018.150.19

[C2] **S. Erniyazov** and J.-C. Jeon. "Half subtractor in QCA for arithmetic circuits." In *Proceedings of the International Workshop on Future Technology (FUTECH 2017)*, Korean Institute of Information Technology, pp. 148–149, July 2017. Proceedings record

### Manuscripts Under Review — Not Included in Publication Count

[M1] **S. Erniyazov** et al. "Fed-ST-MoE: A federated spatio-temporal GNN with mixture-of-experts for heterogeneous building energy and demand forecasting." Manuscript under review.
Investigates decentralized forecasting across heterogeneous buildings with spatio-temporal graph learning and client-aware mixture-of-experts. The study addresses statistical heterogeneity and local-data retention; it does not claim a formal privacy guarantee. Venue and preprint link will be disclosed as permitted by the review policy.

### Exploratory Studies in Progress — External Methodological Collaboration, Not Publications

Thermal and audio sensing serve as methodological stress-test environments for my research on distribution-resilient physical AI. Both studies are in progress; blind or external evaluation remains pending, and no peer-reviewed output is claimed. Collaborating institutions and faculty will be named once public attribution is agreed.

- **ThermalGuard** — exploratory study of identity-minimizing thermal perception for fall-event detection under subject, site, and sensor shift; evaluates event-level sensitivity, false alarms, detection delay, and edge-inference constraints. No formal privacy guarantee is claimed.
- **AudioShield** — exploratory cross-corpus anti-spoofing study using WavLM and XLS-R under codec, resampling, replay, and corpus shift; six controlled experiments and a domain-reliance analysis are public. Blind evaluation pending; no final performance or state-of-the-art claim. Code

## Selected Open Research Artifacts

Public methodological studies, reproducibility instruments, and research prototypes; none is a peer-reviewed publication or an operational deployment. Index: sarvarbek-erniyazov.github.io.

### WaveSeg — Frequency-Boundary Adapter for Medical Image Segmentation

Public code, model weights, and interactive demo. Adds 770 parameters to SegFormer-B0; reported LGG HD95 decreases from 29.70 to 22.83 pixels; 58.27 FPS at 256×256 on an RTX 4060. Includes patient-level splitting, boundary metrics, ablations, and a documented negative auxiliary-loss result. A methodological artifact, not a claim of clinical validation.

Code • Model weights • Interactive demo

### CellTriage — Risk-Controlled Battery Quality Decisions

Public code and interactive research console for early-cycle life prediction, conformal intervals, and cost-aware accept/reject/defer decisions (Extra Trees: MAPE 9.57% $\pm$ 3.35, $\rho = 0.944$; 14.2× cost reduction versus AQL sampling, conditional on the stated cost matrix). Reported conformal coverage falls from about 90% in distribution to 60% under held-out campaign shift, documenting failure of nominal coverage when exchangeability is violated.

Code • Research console

### DriftSentinel — Self-Auditing Distribution-Shift Monitoring

Executable comparison of PSI, KS and chi-square tests with BH-FDR correction, CUSUM, MMD, and BBSD on a 101,766-encounter clinical model. Uses 20-seed negative controls to identify false drift alarms and invalid detector behavior; overturns an initial "8/8 critical" result. Includes 93 automated tests, continuous integration, and determinism checks.

Code • Hosted monitoring console

### FreqFD — Cross-Generator Deepfake Detection

Public dual-stream RGB and frequency-domain detector (19.4M parameters) evaluated under a leakage-audited cross-generator protocol. Reported unseen-generator AUC improves from 0.588 to 0.686; the remaining generalization limitation is explicitly acknowledged.

Code

**Additional public artifacts:** ReproLab (AI4I-2020 reproducibility audit; F1 ceiling 0.9558 derived three ways, spread 0.0015); DALM-DETR (density-adaptive matching, public checkpoints; strongest reported zero-shot transfer, no in-domain superiority claimed); OCRipple (OCR-error propagation study; ANLS 0.848, layout mAP@50 73.8%); FinRiskGuard; Auto Damage Segmentation; Uzbek Handwriting OCR (TrOCR model release).

## Funding Register

| Program / ID | Project and role | Status |
|---|---|---|
| TIPA RS-2026-25529976 | sLM–LMM Cooperative VPP Platform for on-device AI and cloud-integrated energy management. Lead AI Architect / Lead Researcher, assigned university AI work package; not grant PI. | Funded, active, 2026–2028 |
| TIPA RS-2025-02312851 | Preliminary study for an on-device AI and cloud-integrated VPP platform. Lead AI Researcher for ML feasibility, architecture planning, and prototype evaluation. | Funded, completed, 2025 |
| NRF RS-2025-00564485 | Satellite–ground data integration for solar variability management and VPP operations. Methodology contributor. | Submitted proposal |
| INNOPOLIS RS-2026-25550513 | Global validation of AI-based power-equipment anomaly detection and preventive control. Methodology contributor. | Submitted proposal |
| Earlier programs | NRF/MSIP NRF-2017R1D1A3B03034346; MOE Regional Innovation Strategy 2021RIS-002; MSS RS-2023-00278623. Research contributor. | Funded, completed |

## Methods

Spatio-temporal graph learning; multimodal fusion; mixture-of-experts; federated and personalized learning; time-series forecasting; anomaly and drift detection; conformal prediction; uncertainty quantification; selective prediction; computer vision; frequency-domain learning; and edge-aware evaluation.

## Open Science and Academic Service

- Maintainer of public reproducibility artifacts, model cards, and research demonstrations supporting transparent evaluation (leakage audits, negative controls, multi-seed reporting, automated tests, and continuous integration across the artifacts listed above).
- Laboratory service at the AI Laboratory, Chonnam National University: research-pipeline and evaluation-protocol support for national R&D deliverables.